

3-2-1 Backup Rule



3 backup copies



2 on different
media types



1 stored off-site

backup and one copy. It is just the original. If those same photos also exist on an external SSD and in a cloud account, now you have three copies.

The aim is simple. If one copy fails, another exists. If two copies are compromised in the same event, a third still survives.

For a normal reader, this does not mean every file you have ever downloaded deserves three copies forever. Your backup system should protect what would actually hurt to lose: family media, work in progress, IDs, tax records, scans, invoices, project folders, passwords, creative work, and the kinds of documents that are always impossible to locate on the exact day you need them. The 3-2-1 rule is about important data, not every meme you accidentally saved from a WhatsApp group in 2023.

What “2” really means

The second number is two different media types. This part gets ignored most often, because people assume two copies on two drives is enough. Better than one? Absolutely. But the original logic of the rule is that different storage media fail differently. If your only copies are on two devices that live in the same room, use the same power, face the same theft risk, or get hit by the same ransomware infection, you have duplicated your convenience more than your resilience. NIST’s 3-2-1 explanation specifically says two different media types help protect against different hazards.

For consumers, “different media” can be interpreted pragmatically. One copy may live on your laptop or phone, another on an external SSD or hard drive, and another in cloud storage. That is already far more sensible than putting everything on

one internal SSD and hoping Windows, Android, or macOS will somehow remember your intentions when disaster strikes.

What “1” really means

The last number is one off-site copy. This is where the rule stops being merely neat and starts becoming powerful. A second local copy helps if your laptop dies. It does not help much if your home is burgled, a power surge takes out multiple devices. One copy should be somewhere else: cloud storage, a drive stored at another location, or any setup that keeps it physically or logically away from the original device.

For most people in 2026, the easiest off-site copy is cloud storage. Google One in India offers tiers such as 30GB at ₹59/month, 100GB at ₹130/month, and 2TB plans for people who need much more room. Apple’s India pricing for iCloud+ starts at 50GB for ₹75/month, then 200GB at ₹219/month and 2TB at ₹749/month. Microsoft’s OneDrive-linked consumer plans in India include Microsoft 365 Basic with 100GB storage and higher plans like Personal and Family with 1TB per person. These are not just “nice-to-have” subscriptions anymore; they are often the most realistic off-site leg of a 3-2-1 setup.

The simplest way to visualize the rule

Here is the easiest way to think about 3-2-1:

- **Copy 1:** the file where you actually use it
- **Copy 2:** a second local backup
- **Copy 3:** an off-site backup

That is it.

So if you shoot photos on your phone, then copy them to a laptop, and also back them up to Google Photos or iCloud, you